

Untargeted Metabolomics - Materials and Methods

1. Metabolites Extraction

Tissue sample

Tissue samples (100 mg) were individually ground in liquid nitrogen, and the resulting homogenates were resuspended in pre-chilled 80% methanol by vortexing. Samples were incubated on ice for 5 min and centrifuged at $15,000 \times g$ at $4\text{ }^{\circ}\text{C}$ for 20 min. An aliquot of the supernatant was diluted with LC-MS-grade water to a final methanol concentration of 53%. The diluted extract was transferred to a fresh Eppendorf tube and centrifuged again at $15,000 \times g$ at $4\text{ }^{\circ}\text{C}$ for 20 min. The final supernatant was subjected to LC-MS/MS analysis ^[1].

Liquid sample

Liquid samples (100 μL) were transferred into Eppendorf tubes and mixed with pre-chilled 80% methanol by vortexing. After incubation on ice for 5 min, samples were centrifuged at $15,000 \times g$ at $4\text{ }^{\circ}\text{C}$ for 20 min. An aliquot of the supernatant was diluted with LC-MS-grade water to obtain a final methanol concentration of 53%. The diluted extract was transferred to a fresh tube and centrifuged again at $15,000 \times g$ at $4\text{ }^{\circ}\text{C}$ for 20 min. The resulting supernatant was analyzed by LC-MS/MS ^[2-3].

Cell or bacteria sample

Cell or bacterial pellets were resuspended in pre-chilled 80% methanol by vortexing. Samples were thawed on ice and vortexed for 30 s, followed by sonication for 6 min. After centrifugation at 5,000 rpm at $4\text{ }^{\circ}\text{C}$ for 1 min, the supernatant was collected, freeze-dried, and reconstituted in 10% methanol. The reconstituted solution was injected into the LC-MS/MS system for analysis ^[4-5].

Cell or bacteria culture medium sample

Culture medium samples (1 mL) were freeze-dried and resuspended in pre-chilled 80% methanol by vortexing. The samples were incubated on ice for 5 min and centrifuged at $15,000 \times g$ at $4\text{ }^{\circ}\text{C}$ for 15 min. An aliquot of the supernatant was diluted with LC-MS-grade water to a final methanol concentration of 53%. The diluted extract was transferred to a fresh tube and centrifuged again at $15,000 \times g$ at $4\text{ }^{\circ}\text{C}$ for 15 min. The final supernatant was subjected to LC-MS/MS analysis.

Soil sample

Soil samples (1 g) were placed into Eppendorf tubes and resuspended in pre-chilled 80% methanol by vortexing. Samples were incubated on ice for 5 min and centrifuged at $15,000 \times g$ at $4\text{ }^{\circ}\text{C}$ for 15 min. The supernatant was collected and centrifuged again at $15,000 \times g$ at $4\text{ }^{\circ}\text{C}$ for 20 min. The final supernatant was used for LC-MS/MS analysis.

2. UHPLC-MS/MS Analysis (Key Point : Method ② only for Untargeted Metabolomics Pro)

①UHPLC-MS/MS analyses were performed using a Vanquish UHPLC system

(ThermoFisher, Germany) coupled with an Orbitrap Exploris™ 120 mass spectrometer (Thermo Fisher, Germany) in Novogene Co., Ltd. (Cambridge, British). Samples were injected onto a Hypersil Gold column (100×2.1 mm, 1.9µm) using a 12-min linear gradient at a flow rate of 0.2 mL/min. The eluents for the positive and negative polarity modes were eluent A (0.1% FA in Water) and eluent B (Methanol). The solvent gradient was set as follows: 2% B, 1.5 min; 2-85% B, 3 min; 85-100% B, 10 min; 100-2% B, 10.1 min; 2% B, 12 min. Q Exactive™ HF mass spectrometer was operated in positive/negative polarity mode with spray voltage of 3.5 kV, capillary temperature of 320°C, sheath gas flow rate of 35 psi and aux gas flow rate of 10 L/min, S-lens RF level of 60, Aux gas heater temperature of 350°C.

②UHPLC-MS/MS analyses were performed using a Vanquish UHPLC system (ThermoFisher, Germany) coupled with an Orbitrap Exploris™ 120 mass spectrometer (ThermoFisher, Germany) in Novogene Co., Ltd. (Cambridge, British). Samples were injected onto an ACQUITY UPLC BEH Amide Column (100×2.1 mm, 1.7 µm) using a 12-min linear gradient at a flow rate of 0.2 mL/min. The eluents for the positive and negative polarity mode were eluent A (5 mM ammonium acetate in 90% ACN,) and eluent B (5 mM ammonium acetate in 50% ACN,). The solvent gradient was set as follows: 2% B, 1.5 min; 2-100% B, 7 min; 100% B, 9 min; 100-2% B, 9.1 min; 2% B, 12 min. Q Exactive™ HF mass spectrometer was operated in positive/negative polarity mode with spray voltage of 3.5 kV, capillary temperature of 320°C, sheath gas flow rate of 35 arb and aux gas flow rate of 10 arb, S-lens RF level of 60, Aux gas heater temperature of 350°C.

3. Data processing and metabolite identification

The data files generated by UHPLC-MS/MS were processed by XCMS to perform peak alignment, peak picking, and quantitation for each metabolite. Then, based on adduct ions and setting mass deviation to 10 ppm^[6], a comparison was made between these data and the self-built high-quality secondary spectrum database (NovoMetDB) to obtain results for metabolite identification. Metabolites with more than 50% missing values in the samples were first filtered out and the remaining missing values were imputed using KNN^[7]. After eliminating background ions according to blank samples, the raw quantitative results were further processed to obtain relative peak areas, which included normalization and filtering out compounds whose relative peak areas exhibited a coefficient of variation (CV) greater than 30% in the quality control (QC) samples^[8]. Relative peak areas = Raw quantitative value of samples/ (The sum of quantitative value of samples/ The sum of quantitative value of QC1). Data processing is based on the Linux operating system (CentOS version 6.6), using R and Python. For details on specific packages and software versions used, please refer to the README file included in the results.

4. Data Analysis

These metabolites were annotated using the KEGG database (<https://www.genome.jp/kegg/pathway.html>) , HMDB database (<https://hmdb.ca/metabolites>) and LIPIDMaps database (<http://www.lipidmaps.org/>). Principal components analysis (PCA) and Partial least squares discriminant analysis (PLS-DA) were performed at metaX^[9] (a flexible and comprehensive software for processing metabolomics data). We

applied univariate analysis (t-test) to calculate the statistical significance (P-value). FC refers to the fold change, which is the ratio of the mean of all biological repeated quantitative values of each metabolite in the comparison group. The differential metabolites were screened using VIP, P-value and FC. The specific standard outlined in Table 5.3 of the report. Volcano plots were used to filter metabolites of interest which based on $\log_2(\text{FoldChange})$ and $-\log_{10}(\text{p-value})$ of metabolites by ggplot2 in R language.

For clustering heat maps, the data were normalized using z-scores of the intensity areas of differential metabolites and were plotted by Pheatmap package in R language. The correlation between differential metabolites were analyzed by cor () in R language (method=pearson). Statistically significant of correlation between differential metabolites were calculated by cor.mtest () in R language. P-value < 0.05 was considered as statistically significant and correlation plots were plotted by corrplot package in R language. The functions of these metabolites and metabolic pathways were studied using the KEGG database. The metabolic pathways enrichment of differential metabolites was performed, when ratio were satisfied by $x/n > y/N$, metabolic pathway were considered as enrichment, when P-value of metabolic pathway < 0.05, metabolic pathway were considered as statistically significant enrichment.

Reference

- [1] Want E J , Masson P , Michopoulos F , et al. Global metabolic profiling of animal and human tissues via UPLC-MS[J]. Nature Protocols, 2012, 8(1):17-32.
- [2] Want E J , O'Maille G , Smith C A , et al. Solvent-Dependent Metabolite Distribution, Clustering, and Protein Extraction for Serum Profiling with Mass Spectrometry[J]. Analytical Chemistry, 2006, 78(3):743-752.
- [3] Barri T , Dragsted L O . UPLC-ESI-QTOF/MS and multivariate data analysis for blood plasma and serum metabolomics: effect of experimental artefacts and anticoagulant[J]. Analytica Chimica Acta, 2013, 768(1):118-128.
- [4] Sellick C A , Hansen R , Stephens G M , et al. Metabolite extraction from suspension cultured mammalian cells for global metabolite profiling[J]. Nature Protocol, 2011, 6(8):1241-9.
- [5] Yuan M , Breitkopf S B , Yang X , et al. A positive/negative ion-switching, targeted mass spectrometry-based metabolomics platform for bodily fluids, cells, and fresh and fixed tissue[J]. Nature Protocols, 2012, 7(5):872-81.
- [6] Li, Yuanyue et al. Spectral entropy outperforms MS/MS dot product similarity for small-molecule compound identification[J]. Nature methods, 2021, vol. 18,12: 1524-1531.
- [7] Do KT, Wahl S, Raffler J, et al. Characterization of missing values in untargeted MS-based metabolomics data and evaluation of missing data handling strategies[J]. Metabolomics. 2018;14(10):128.
- [8] Dai, W. et al. Characterization of white tea metabolome: Comparison against green and black tea by a nontargeted metabolomics approach[J]. Food Research International, 2017, 96,40–45.
- [9] Wen B , Mei Z , Zeng C , et al. metaX: a flexible and comprehensive software for processing metabolomics data[J]. BMC Bioinformatics, 2017, 18(1).